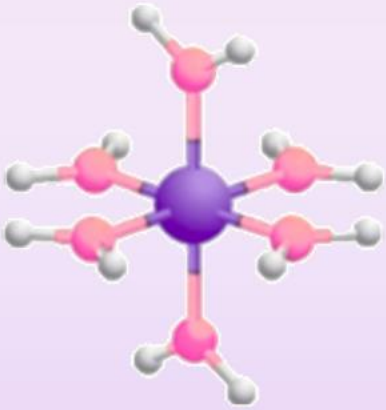
CHEM 1043 General Chemistry II

Topic 4

Solubility and Complex-Ion Equilibria



Dr. A T L LI

What to learn in this topic?



Solubility and Complex-Ion Equilibria

- ✓ Solubility product constant, K_{sp}
- ✓ Relationship between solubility and K_{sp}
- ✓ Common-ion effect in solubility equilibria
- ✓ Criteria for precipitation and its completeness
- ✓ Fractional precipitation
- ✓ Solubility and pH
- ✓ Equilibria involving complex ions
- ✓ Qualitative cation analysis

Your prior knowledge related to this topic

HKDSE Chemistry:

Topic X Chemical Equilibrium

Students should learn	Students should be able to
a. Dynamic equilibrium • characteristics of dynamic equilibrium	• describe reversible and irreversible reactions by using suitable examples • describe characteristics of a system existing in dynamic equilibrium
b. Equilibrium constant • equilibrium constant expressed in terms of concentrations (K_c)	• express the mathematical relationship between concentrations of reactants and products at equilibrium and K_c
c. The effect of changes in concentration and temperature on chemical equilibria • a change in temperature results in possible changes in K_c of the system • changes in concentration result in the adjustment of the system without changing the value of K_c	• recognise that the value of K_c for an equilibrium system is a constant at constant temperature irrespective of changes in concentration of reactants and products • perform calculations involving K_c • perform practical work on the determination of K_c • derive inductively the relation of temperature and the value of K_c from given data sets • predict qualitatively the effect of temperature on the position of equilibrium from the sign of ΔH for the forward reaction • deduce the effect of change in concentration on the position of chemical equilibrium

Your prior knowledge related to this topic

CHEM1042 General Chemistry I

Topic 5. Acid-Base Equilibria

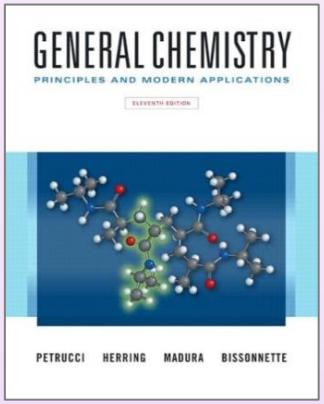
- ✓ Theories describing acids and bases
- ✓ Self-ionisation of water and the pH scale
- ✓ Strength of acids and bases
- ✓ Basicity and polyprotic acids
- ✓ Acid-base reactions and neutralisation
- ✓ Acid-base properties of salt solutions
- ✓ Molecular structure and acid-base behaviour
- ✓ Common-ion effect in acid-base equilibria
- ✓ Buffer solutions
- ✓ Acid-base indicators
- ✓ Volumetric analysis involving acids and bases



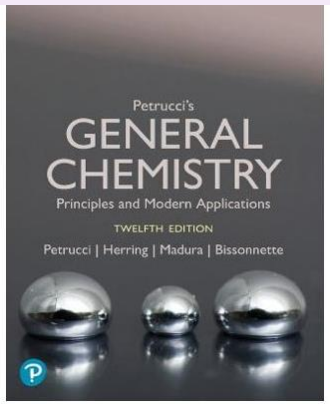
Your reading:

Petrucci; Herring; Madura; Bissonnette:
General Chemistry: Principles and Modern Applications, 11th or 12th edition, Pearson

Chapter 18) Solubility and Complex-Ion Equilibria



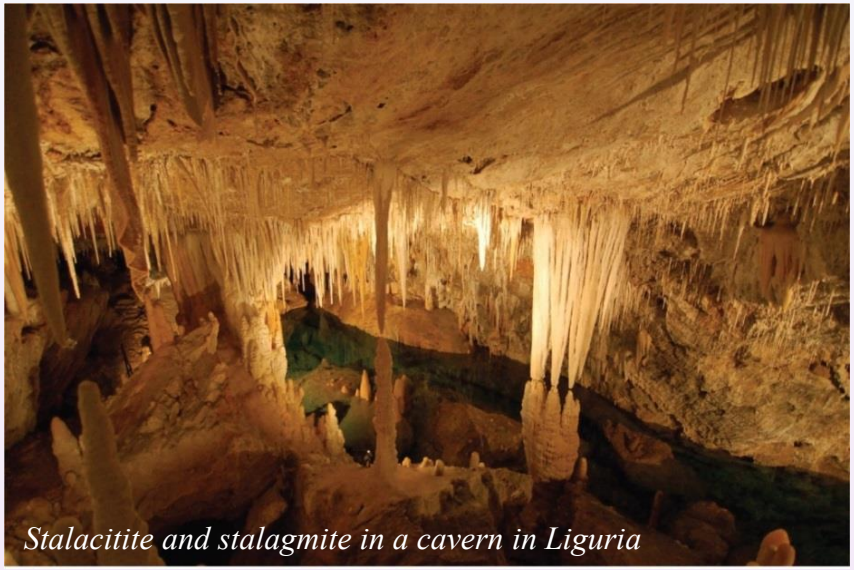
11th edition



12th edition

Think About:

1.



Stalacitite and stalagmite in a cavern in Liguria

Stalactite and stalagmite are formed from dissolution and precipitation of limestone ($CaCO_3$). What are the conditions for $CaCO_3(s)$ to dissolve and those for a solution containing Ca^{2+} and CO_3^{2-} ions to precipitate?

2.



$AgNO_3(aq) + NaCl(aq)$

$\rightarrow$ **AgCl(s)** $+NaNO_3(aq)$

No AgCl will precipitate if the solution contains a moderate to high conc. of $NH_3(aq)$. Why?

Solubility Product Constant K_{sp}

- The solubility product constant, K_{sp}, is the constant for the equilibrium established between a solid solute and its ions in a saturated solution.

- Consider **Gypsum ($CaSO_4 \cdot 2H_2O$)** which is slightly soluble in water,

$$CaSO_4(s) \rightleftharpoons Ca^{2+}(aq) + SO_4^{2-}(aq)$$

the equilibrium constant expression is $$K = \frac{a_{Ca^{2+}} a_{SO_4^{2-}}}{a_{CaSO_4}}$$

- The activity of pure solid is 1. ∵ the conc. of the ions are very small, their activities can be taken as their molar concentrations. Therefore, the above K expression can be rewritten as:

$$K_{sp} = [Ca^{2+}][SO_4^{2-}] = 9.1 \times 10^{-6} \text{ (at 25 °C)}$$

- Solubility product constant varies with temperature.

Solubility Product Constant K_{sp}

Several Examples of Solubility Product Constants at 25 °C

Solute	Solubility Equilibrium	K_{sp}
Aluminum hydroxide	$Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq) + 3\,OH^-(aq)$	1.3×10^{-33}
Barium carbonate	$BaCO_3(s) \rightleftharpoons Ba^{2+}(aq) + CO_3^{2-}(aq)$	5.1×10^{-9}
Barium sulfate	$BaSO_4(s) \rightleftharpoons Ba^{2+}(aq) + SO_4^{2-}(aq)$	1.1×10^{-10}
Calcium carbonate	$CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$	2.8×10^{-9}
Calcium fluoride	$CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2\,F^-(aq)$	5.3×10^{-9}
Calcium sulfate	$CaSO_4(s) \rightleftharpoons Ca^{2+}(aq) + SO_4^{2-}(aq)$	9.1×10^{-6}
Chromium(III) hydroxide	$Cr(OH)_3(s) \rightleftharpoons Cr^{3+}(aq) + 3\,OH^-(aq)$	6.3×10^{-31}
Iron(III) hydroxide	$Fe(OH)_3(s) \rightleftharpoons Fe^{3+}(aq) + 3\,OH^-(aq)$	4×10^{-38}
Lead(II) chloride	$PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,Cl^-(aq)$	1.6×10^{-5}
Lead(II) chromate	$PbCrO_4(s) \rightleftharpoons Pb^{2+}(aq) + CrO_4^{2-}(aq)$	2.8×10^{-13}
Lead(II) iodide	$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,I^-(aq)$	7.1×10^{-9}
Magnesium carbonate	$MgCO_3(s) \rightleftharpoons Mg^{2+}(aq) + CO_3^{2-}(aq)$	3.5×10^{-8}
Magnesium fluoride	$MgF_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\,F^-(aq)$	3.7×10^{-8}
Magnesium hydroxide	$Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\,OH^-(aq)$	1.8×10^{-11}
Magnesium phosphate	$Mg_3(PO_4)_2(s) \rightleftharpoons 3\,Mg^{2+}(aq) + 2\,PO_4^{3-}(aq)$	1×10^{-25}
Mercury(I) chloride	$Hg_2Cl_2(s) \rightleftharpoons Hg_2^{2+}(aq) + 2\,Cl^-(aq)$	1.3×10^{-18}
Silver bromide	$AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq)$	5.0×10^{-13}
Silver carbonate	$Ag_2CO_3(s) \rightleftharpoons 2\,Ag^+(aq) + CO_3^{2-}(aq)$	8.5×10^{-12}
Silver chloride	$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$	1.8×10^{-10}
Silver chromate	$Ag_2CrO_4(s) \rightleftharpoons 2\,Ag^+(aq) + CrO_4^{2-}(aq)$	1.1×10^{-12}
Silver iodide	$AgI(s) \rightleftharpoons Ag^+(aq) + I^-(aq)$	8.5×10^{-17}
Strontium carbonate	$SrCO_3(s) \rightleftharpoons Sr^{2+}(aq) + CO_3^{2-}(aq)$	1.1×10^{-10}
Strontium sulfate	$SrSO_4(s) \rightleftharpoons Sr^{2+}(aq) + SO_4^{2-}(aq)$	3.2×10^{-7}



Class Practice 1

Solubility Product Constant Expression for Slightly Soluble Solutes

Write the solubility product constant expression for the solubility equilibrium of

(a) Calcium fluoride, CaF_2 (one of the substances used when a fluoride treatment is applied to teeth).

(b) Copper arsenate, $Cu_3(AsO_4)_2$ (used as an insecticide and fungicide).

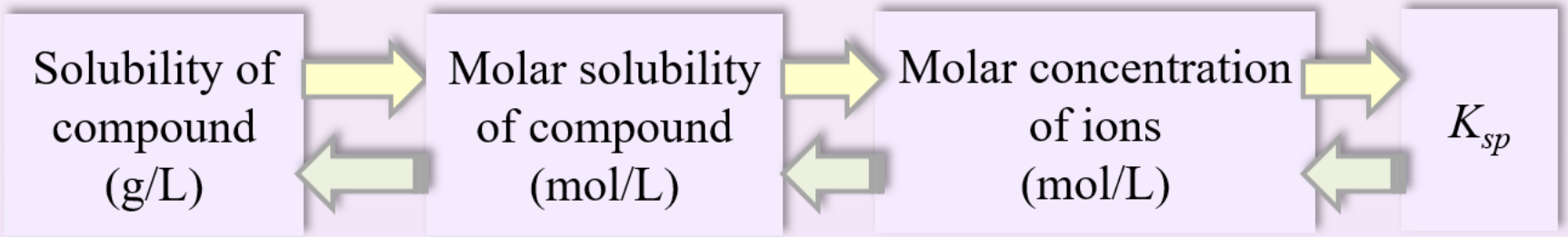
Relationship between Solubility and K_{sp}

- There is a relationship between the solubility product constant K_{sp} of a solute and the solute's molar solubility – its molarity in a saturated aqueous solution.

$$AB\ (s) \rightleftharpoons A^+(aq) + B^-(aq)$$

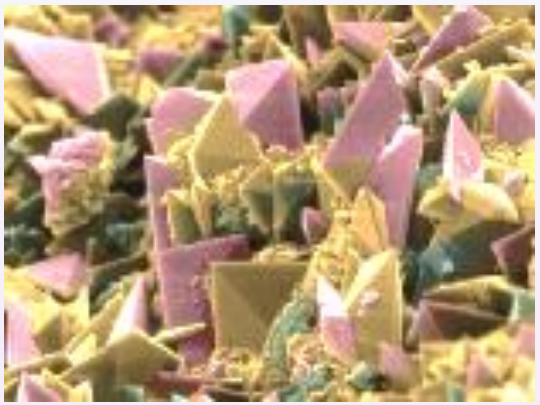
$$K_{sp} = [A^+][B^-]$$

Take-home message:



Relationship between Solubility and K_{sp}

Applications:



Example 1: Calcium oxalate (CaC_2O_4) ($K_{sp} = 2.7 \times 10^{-9}$)

Scanning electron microscope (SEM) image of calcium oxalate crystals, a common type of kidney stone that can form in the human kidney.

Example 2: Barium sulphate ($BaSO_4$) ($K_{sp} = 1.1 \times 10^{-10}$)

Barium sulphate is a radiographic contrast agent indicated for use in computed tomography (CT) of the GI tract. It is a good absorber of X-rays. As a component of a "barium milk shake," it coats the intestinal tract so that this soft tissue will show up when X-rayed. Even though Ba^{2+} is poisonous, it is harmless because its aqueous solubility is very low.



Example 1

Calculating K_{sp} of a Slightly Soluble Solute from Its Solubility

A handbook lists the aqueous solubility of $CaSO_4$ at 25 °C as 0.20 g / 100 mL. What is the K_{sp} of $CaSO_4$ at 25 °C?

$$CaSO_4(s) \rightleftharpoons Ca^{2+}(aq) + SO_4^{2-}(aq) \qquad K_{sp} = ?$$

Suggested Solution:

$$\text{g } CaSO_4/100 \text{ mL} \longrightarrow \text{mol } CaSO_4/\text{L} \longrightarrow [Ca^{2+}] \text{ and } [SO_4^{2-}] \longrightarrow K_{sp}$$

$$\frac{\text{mol } CaSO_4}{\text{L satd soln}} = \frac{0.20 \text{ g } CaSO_4}{0.100 \text{ L soln}} \times \frac{1 \text{ mol } CaSO_4}{136 \text{ g } CaSO_4}$$

$$= 0.015 \text{ M } CaSO_4$$

Using stoichiometric ratios (shown in blue),

$$[Ca^{2+}] = \frac{0.015 \text{ mol } CaSO_4}{1 \text{ L}} \times \frac{1 \text{ mol } Ca^{2+}}{1 \text{ mol } CaSO_4} = 0.015 \text{ M}$$

$$[SO_4^{2-}] = \frac{0.015 \text{ mol } CaSO_4}{1 \text{ L}} \times \frac{1 \text{ mol } SO_4^{2-}}{1 \text{ mol } CaSO_4} = 0.015 \text{ M}$$

$$K_{sp} = [Ca^{2+}][SO_4^{2-}] = (0.015)(0.015) = 2.3 \times 10^{-4}$$

Example 2

Calculating the Solubility of a Slightly Soluble Solute from its K_{sp} value

Lead(II) iodide, PbI_2, is a dense, golden yellow, "insoluble" solid used in bronzing and in ornamental work requiring a golden color (such as mosaic gold). Calculate the molar solubility of lead(II) iodide in water at 25 °C, given that $K_{sp} = 7.1 \times 10^{-9}$

Suggested Solution:

The solubility equilibrium equation $PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,I^-(aq)$

let s represent the number of moles of PbI_2 dissolved per liter of saturated solution

$$[Pb^{2+}] = s \quad \text{and} \quad [I^-] = 2s$$

$$K_{sp} = [Pb^{2+}][I^-]^2 = (s)(2s)^2 = 7.1 \times 10^{-9}$$
$$4s^3 = 7.1 \times 10^{-9}$$
$$s^3 = 1.8 \times 10^{-9}$$
$$s = (1.8 \times 10^{-9})^{1/3} = 1.2 \times 10^{-3}$$

The molar solubility of PbI_2 in water is 1.2×10^{-3} M.



Class Practice 2

Calculating K_{sp} of a Slightly Soluble Solute from Its Solubility

A handbook lists the aqueous solubility of AgOCN as 7 mg/100 mL at 20 °C. What is the K_{sp} of AgOCN at 20 °C?



Class Practice 3

Calculating the Solubility of a Slightly Soluble Solute from Its K_{sp} Value

The K_{sp} of $Cu_3(AsO_4)_2$ at 25 °C is 7.6×10^{-36}. What is the molar solubility of $Cu_3(AsO_4)_2$ in H_2O at 25 °C?

Solubility Product Constant K_{sp}

Concept Check:

In comparing several salts at a given temperature, does a **higher K_{sp} value** always mean a **higher solubility**?

Let's calculate:

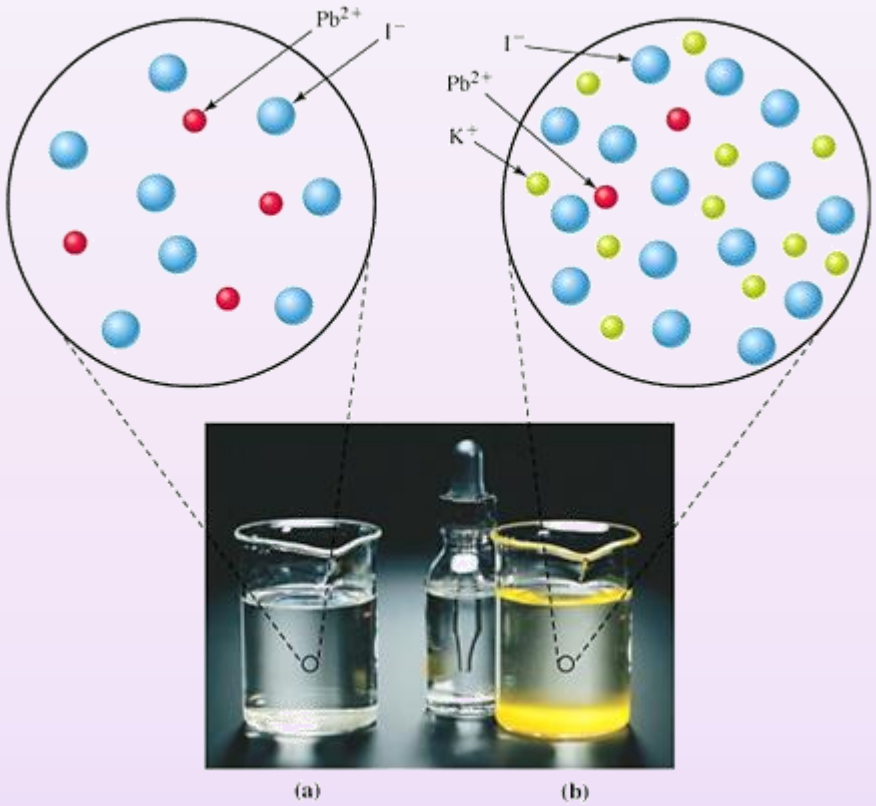
1. The solubility of silver chloride in water. $K_{sp} = 1.6 \times 10^{-10}$

2. The solubility of silver phosphate in water. $K_{sp} = 1.8 \times 10^{-18}$

Common-ion Effect in Solubility Equilibria

❖ Consider PbI in water, when adding some I^-, a common ion, from a source such as KI (aq) to the saturated PbI_2 (aq), the situation is similar to that in **Acid-Base Equilibria**.

$$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,I^-(aq)$$



The common-ion effect in solubility equilibrium

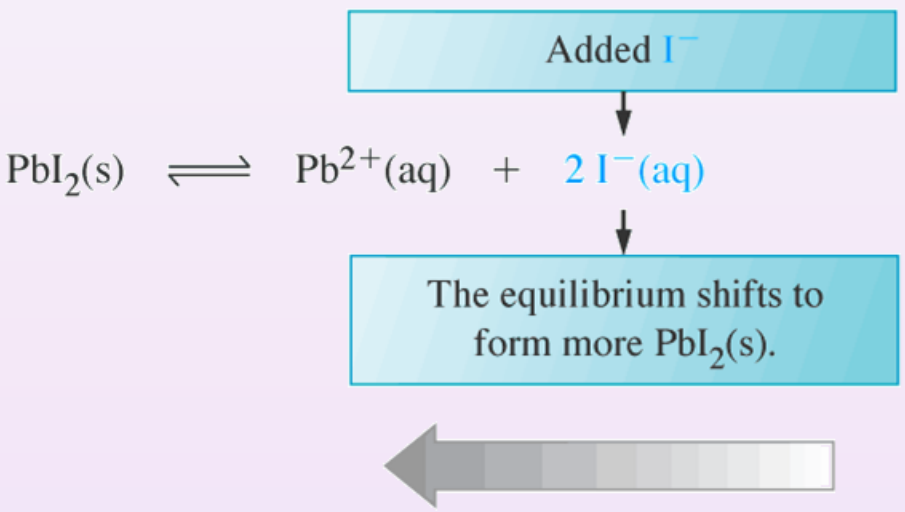
(a) A clear saturated solution of PbI_2 from which excess undissolved solute has been filtered off.

(b) When a small volume of a concentrated solution of KI (containing the common ion, I^-) is added, a small quantity of $PbI_2(s)$ precipitates. **A common ion *reduces the solubility of a sparingly soluble solute.***

Common-ion Effect in Solubility Equilibria

❖ According to Le Châtelier's Principle, when an equilibrium system is subjected to a change in temperature, pressure, or concentration of a reacting species, the system responds by attaining a **new** equilibrium that ***partially*** **offsets** the impact of the change.

∴ An equilibrium mixture responds to a forced increase in the concentration of one of its reactants by shifting in the direction in which that reactant is consumed.



- The addition of common ion I^- into the solubility equilibrium causes the reverse reaction to be favoured.
- The equilibrium shifts toward the undissolved compound, causing more precipitation.
- Thus, **the solubility of the compound is reduced**.

Take-home message:

❖ The solubility of a slightly soluble ionic compound is ***reduced*** in the presence of a second solute that supplies a common ion.

Example 3

Calculating the Solubility of a Slightly Soluble Solute in the Presence of a Common Ion

What is the molar solubility of lead (II) iodide (PbI_2) in 0.10 M KI (aq)?
(Given that $K_{sp} = 7.1 \times 10^{-9}$)

Suggested Solution:

The ICE table is

$$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,I^-(aq)$$

initial concns, M:	0	0.10
from PbI_2, M:	s	$2s$
equil concns, M:	s	$(0.10 + 2s)$

$$K_{sp} = [Pb^{2+}][I^-]^2 = (s)(0.10 + 2s)^2 = 7.1 \times 10^{-9}$$

To simplify the solution to this equation, let's assume that $2s$ is much smaller than 0.10 M, so $0.10 + 2s \approx 0.10$.

$$s(0.10)^2 = 7.1 \times 10^{-9}$$

$$s = \frac{7.1 \times 10^{-9}}{(0.10)^2} = 7.1 \times 10^{-7}$$

The molar solubility of PbI_2 in 0.10 M KI(aq) is 7.1×10^{-7} M.

NOTE: As shown in Examples 2 and 3, the solubility of PbI_2 in the presence of 0.10 M I^- is about 2000 times less than its value in pure water.



Class Practice 4

Calculating the Solubility of a Slightly Soluble Solute in the Presence of a Common Ion

What is the molar solubility of PbI_2 in 0.10 M $Pb(NO_3)_2$ (aq)?
(Given that $K_{sp} = 7.1 \times 10^{-9}$)
[*Hint:* To which ion concentration should the solubility be related?]

Limitations of the K_{sp} Concept

1. K_{sp} in soluble ionic compounds

❖ For moderately or highly soluble ionic compounds (e.g. NaCl, KNO_3 and NaOH), the K_{sp} must be based on **ion activities** rather than on concentrations.

(***Reason***: in ionic solutions of moderate to high conc., activities and concentrations are generally far from equal.)

Note that :

- The difference between stoichiometric concentration and activity (effective concentration) increases as more solute is dissolved in the solvent.
- There are no interactions between solute particles in an ideal solution, stoichiometric and effective concentrations are equal.

Limitation 1:

Thus, K_{sp} values are usually limited to slightly soluble (essentially insoluble) solutes, and ion molarities are used in place of activities.

Limitations of the K_{sp} Concept

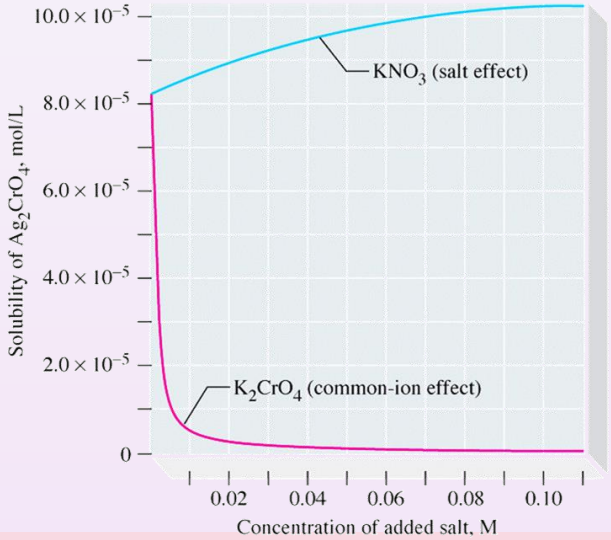
2. The Diverse Non-common Ion Effect: the Salt Effect

❖ The effect of non-common or diverse ions is not as striking as the common-ion effect. The diverse ion effect is commonly called the ***salt effect***.

❖ Diverse ions tend to ***increase*** solubility. As the total ionic concentration of a solution ↑, interionic attractions become more important.

Comparison of the common-ion effect and the salt effect on the molar solubility of Ag_2CrO_4

The presence of CrO_4^{2-} ions, derived from $K_2CrO_4(aq)$, reduces the solubility of Ag_2CrO_4 by a factor of about 35 over the concentration range shown (from 0 to 0.10 M added salt). Over the same concentration range, the solubility of Ag_2CrO_4 is increased by the presence of the diverse ions from KNO_3, but only by about 25%.



Limitation 2:

As a result of the salt effect, the numerical value of a K_{sp} based on molarities will vary depending on the ionic atmosphere.

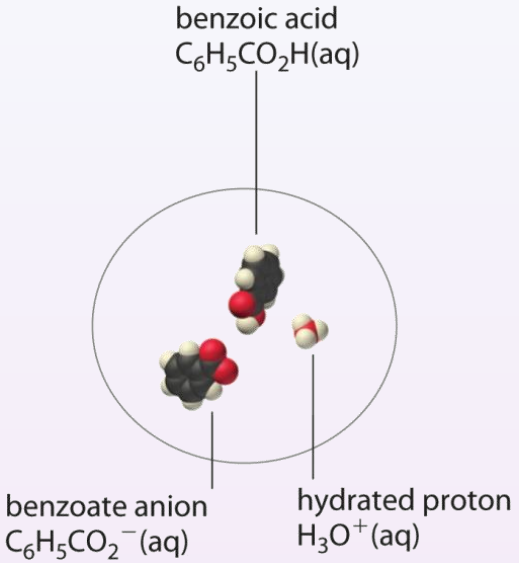
∴ Most tabulated values of K_{sp} are based on activities rather than on molarities.

Limitations of the K_{sp} Concept

3. Incomplete Dissociation of Solute into Ions

Incomplete dissociation

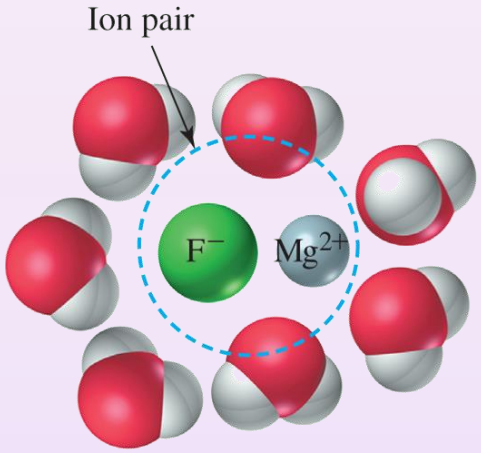
❖ The assumption that all dissolved solute appears in solution as separated cations and anions is often not valid, i.e. the solute may not be 100% ionic, some may be present in the solution in *molecular* form.



Ion-pair formation

❖ Some ions in solution might join together as ***ion pairs*** due to electrostatic attraction.

e.g. In a saturated solution of MgF_2, although most of the solute exists as Mg^{2+} and F^- ions, some exists as the ion pair MgF^+.



∴ the concentrations of the dissociated ions are ***reduced*** from the stoichiometric expectation.

Limitations of the K_{sp} Concept

Limitation 3:

As a result of the ion-pair formation, additional solute must be present for the product of ion concentrations to equal K_{sp} of the solute, making the true solubility of the solute ***greater*** than expected.

❖ Ion-pair formation increases as mutual electrostatic attraction of the anion and cation increases, i.e. multiple charges ions.

4. Simultaneous Equilibria

❖ In addition to the reversible reaction between a solid solute and its ions in aqueous solution, the self-ionization of water also occurs (generally ignored).

❖ Other equilibrium processes may also occur between solute ions and other solution species.

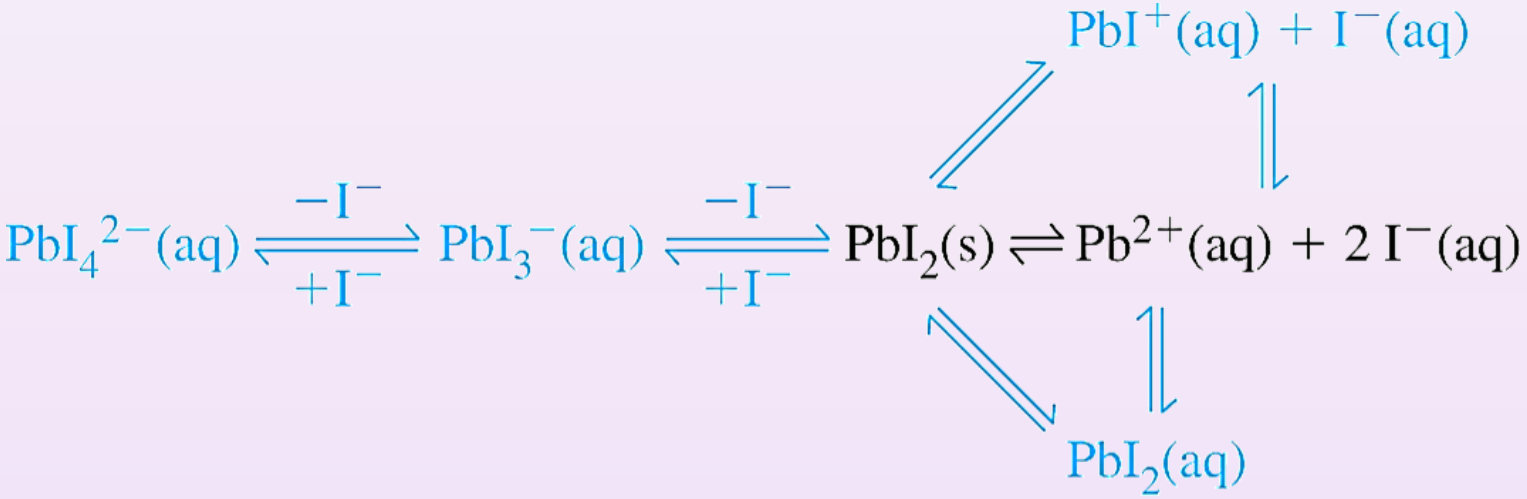
e.g. **acid-base reaction and complex-ion formation**

Limitations of the K_{sp} Concept

Limitation 4:

Calculation based on the K_{sp} expression may bear error if we do not take into account other equilibrium processes that occur simultaneously with solution equilibrium.

- Consider the dissolution process of $PbI_2(s)$ again (see below), it is in fact more complex than we have thought.



There are many competing processes in a dissolution process.

Assessing the Limitation of K_{sp}

- To assess the importance of the effects discussed, we can recall Example 1 in which some of the effects apply. The K_{sp} calculated for $CaSO_4$ on the basis of its measured solubility was 2.3×10^{-4}, which is about 25 times larger than 9.1×10^{-6}.

- The difference is due to the fact that the K_{sp} value listed in slide 9 is based on ion activities, whereas the K_{sp} value calculated from the experimentally determined solubility is based on ion concentrations, assuming complete dissociation of the solute into ions and no ion-pair formation.

- We will continue to **substitute molarities for activities of ions**, although this may not lead to highly accurate result. The general magnitude (within a factor of 10 or 100) still allows us to make some correct predictions and to apply the K_{sp} concept in useful ways.

Criteria for Precipitation and Its Completeness

- Consider the solubility equilibrium of silver iodide and its K_{sp},

$$AgI(s) \rightleftharpoons Ag^+(aq) + I^-(aq)$$
$$K_{sp} = [Ag^+][I^-] = 8.5 \times 10^{-17}$$

- Suppose mixing solutions of $AgNO_3(aq)$ and KI(aq) to obtain a mixed solution that has $[Ag^+] = 0.010$ M and $[I^-] = 0.015$ M. ***Is this solution unsaturated, saturated, or supersaturated?***

- Recall the **reaction quotient, *Q***, it has the same form as an equilibrium constant expression but uses initial concentrations rather than equilibrium concentrations. Initially,

$$Q_{sp} = [Ag^+]_{init} \times [I^-]_{init} = (0.010)(0.015) = 1.5 \times 10^{-4} > K_{sp}$$

- When applied to solubility equilibria, $\boldsymbol{Q_{sp}}$ is generally called the **ion product**.

Criteria for Precipitation and Its Completeness

$$AgI(s) \rightleftharpoons Ag^+(aq) + I^-(aq)$$

$$Q_{sp} = [Ag^+]_{init} \times [I^-]_{init} = (0.010)(0.015) = 1.5 \times 10^{-4} > K_{sp} = 8.5 \times 10^{-17}$$

- If $Q_{sp} > K_{sp}$, $[Ag^+]$ and $[I^-]$ are higher than they would be in a saturated solution and that a net change should occur to the ***left***. The solution is ***supersaturated***, therefore excess AgI should precipitate from solution.

- If $Q_{sp} < K_{sp}$, the solution would have been ***unsaturated***. No precipitate would form.

- To check if precipitation will occur or not, compare Q_{sp} with K_{sp}.

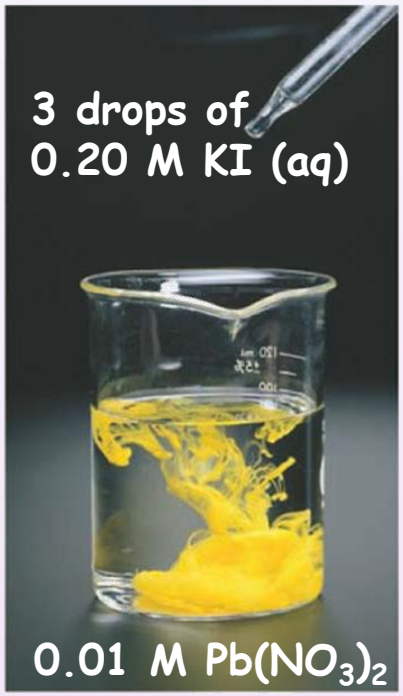
If $Q_{sp} > K_{sp}$, precipitation should occur.

If $Q_{sp} < K_{sp}$, precipitation cannot occur.

If $Q_{sp} = K_{sp}$, a solution is just saturated.

Criteria for Precipitation and Its Completeness

Applying the criteria for precipitation from solution



(a)


(b)

(a) When three drops of 0.20 M KI are first added to 100.0 mL of 0.01 M $Pb(NO_3)_2$, a precipitate forms because K_{sp} is exceeded in the immediate vicinity of the drops.

(b) When the KI becomes uniformly mixed in $Pb(NO_3)_2$ (aq), K_{sp} is no longer exceeded and the precipitate redissolves. The criteria for precipitation must be applied after dilution has occurred.

❖ As a rule of thumb, **precipitation is complete** if **99.9% or more** of a particular ion has precipitated, leaving less than 0.1% of the ion in solution.

Example 4

Applying the Criteria for Precipitation of a Slightly Soluble Solute

Three drops of 0.20 M KI are added to 100.0 mL of 0.01 M $Pb(NO_3)_2$. Will a precipitate of lead(II) iodide form? (Assume 1 drop = 0.05 mL)

$$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,I^-(aq) \qquad K_{sp} = 7.1 \times 10^{-9}$$

Suggested Solution:

Dilution Calculation:

$$\text{amount } I^- = 3 \text{ drops} \times \frac{0.05 \text{ mL}}{1 \text{ drop}} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{0.20 \text{ mol KI}}{\text{L}} \times \frac{1 \text{ mol } I^-}{1 \text{ mol KI}} = 3 \times 10^{-5} \text{ mol } I^-$$

$$\text{Total volume} = 0.1000 \text{ L} + \left(3 \text{ drops} \times \frac{0.05 \text{ mL}}{\text{drop}} \times \frac{1 \text{ L}}{1000 \text{ mL}}\right) = 0.1002 \text{ L}$$

$$[I^-] = \frac{3 \times 10^{-5} \text{ mol } I^-}{0.1002 \text{ L}} = 3 \times 10^{-4} \text{ M}$$

Applying Precipitation Criteria:

$$Q_{sp} = [Pb^{2+}][I^-]^2 = (0.010)(3 \times 10^{-4})^2 = 9 \times 10^{-10}$$

Because a Q_{sp} of 9×10^{-10} is *smaller than* a K_{sp} of 7.1×10^{-9}, we conclude that $PbI_2(s)$ should *not* precipitate.



Class Practice 5

Applying the Criteria for Precipitation of a Slightly Soluble Solute

Three drops of 0.20 M KI are added to 100.0 mL of a 0.010 M solution of $AgNO_3$. Will a precipitate of silver iodide form?

(Assume 1 drop = 0.05 mL, given that K_{sp} of AgI at 25 °C = 8.5×10^{-17}.)

Example 5

Assessing the Completeness of a Precipitation Reaction

The first step in a commercial process in which magnesium is obtained from seawater involves precipitating Mg^{2+} as $Mg(OH)_2$ (s). The magnesium ion concentration in seawater is about 0.059 M. If a seawater sample is treated so that its $[OH^-]$ is maintained at 2.0×10^{-3} M,

(a) what will be $[Mg^{2+}]$ remaining in solution when precipitation stops?

(Given that K_{sp} of $Mg(OH)_2$ at 25 °C = 1.8×10^{-11})

(b) Is the precipitation of $Mg(OH)_2$ (s) complete under these conditions?

conditions?

Suggested Solution:

(a) There is no question that precipitation will occur, because the ion product $Q_{sp} = [Mg^{2+}][OH^-]^2 = (0.059)(2.0 \times 10^{-3})^2 = 2.4 \times 10^{-7}$ exceeds K_{sp}.

$$[Mg^{2+}][OH^-]^2 = [Mg^{2+}](2.0 \times 10^{-3})^2 = 1.8 \times 10^{-11} = K_{sp}$$

$$[Mg^{2+}]_{remaining} = \frac{1.8 \times 10^{-11}}{(2.0 \times 10^{-3})^2} = 4.5 \times 10^{-6}\ M$$

(b) $[Mg^{2+}]$ in seawater is reduced from 0.059 M to 4.5×10^{-6} M as a result of the precipitation reaction. Expressed as a percentage,

$$\%[Mg^{2+}]_{remaining} = \frac{4.5 \times 10^{-6}\ M}{0.059\ M} \times 100\% = 0.0076\%$$

Because less than 0.1% of the Mg^{2+} remains, we conclude that precipitation is essentially complete.



Class Practice 6

Assessing the Completeness of a Precipitation Reaction

A typical Ca^{2+} concentration in seawater is 0.010 M. Will the precipitation of $Ca(OH)_2$ be complete from a seawater sample in which $[OH^-]$ is maintained at 0.040 M? Given K_{sp} of $Ca(OH)_2$ at 25 °C = 5.5×10^{-6}.

Fractional Precipitation

- ***Fractional precipitation*** is a technique in which two or more ions in solution, each capable of being precipitated by the same reagent, are separated by the proper use of that reagent: *One ion is precipitated, while the other(s) remains in solution.*

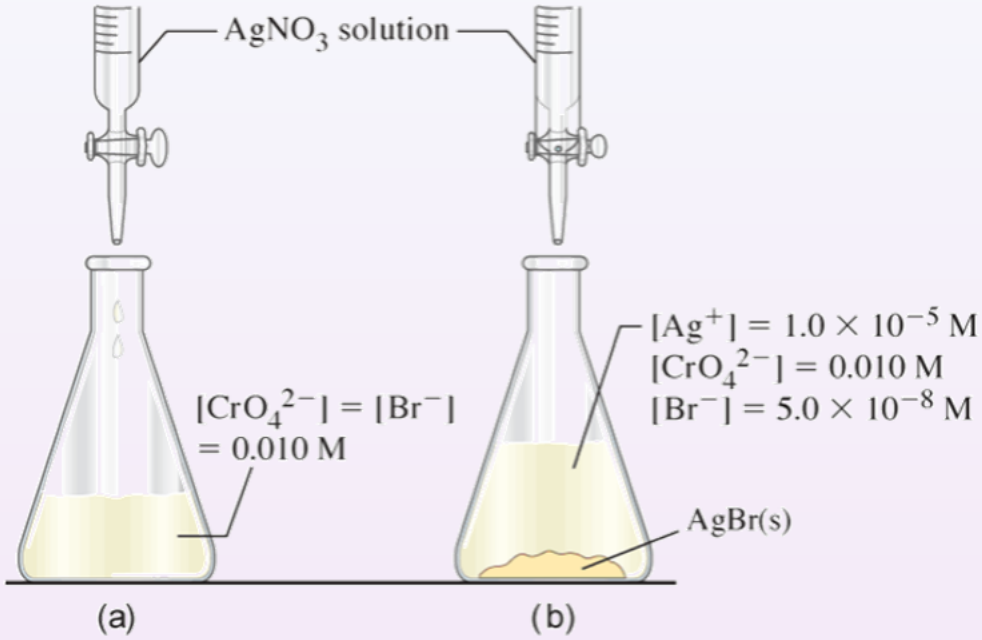
- Consider to a solution containing the ions CrO_4^{2-} and Br^-, a large excess of $AgNO_3(s)$ is added, a mixed precipitate of $Ag_2CrO_4(s)$ and $AgBr(s)$ is obtained. There is a method of adding $AgNO_3$ so as to cause $AgBr(s)$ to precipitate but leave CrO_4^{2-} in solution.

- Criteria for a successful fractional precipitation:

1. a **significant difference in the solubilities** (usually means the K_{sp} values) of the substances being separated.

2. the **addition** of a concentrated solution of the precipitating reagent to the solution from which precipitation is to occur must be **slow**.

Fractional Precipitation



$$Ag_2CrO_4(s) \rightleftharpoons 2\,Ag^+(aq) + CrO_4^{2-}(aq) \qquad K_{sp} = 1.1 \times 10^{-12}$$

$$AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq) \qquad K_{sp} = 5.0 \times 10^{-13}$$

Example:

(a) $AgNO_3(aq)$ is ***slowly*** added to a solution that is 0.010 M in Br^- and 0.010 M in CrO_4^{2-}.

(b) Essentially all the Br^- has precipitated as pale yellow AgBr(s),with $[Br^-]$ in solution = 5.0×10^{-8} M. Red-brown $Ag_2CrO_4(s)$ is just about to precipitate.

Example 6

Separating Ions by Fractional Precipitation

$AgNO_3$ (aq) is slowly added to a solution that has 0.010 M [CrO_4^{2-}] and 0.010 M [Br^-].

(a) Show that AgBr (s) should precipitate before Ag_2CrO_4 (s) does.

(b) When Ag_2CrO_4 (s) begins to precipitate, what is [Br^-] remaining in solution?

(c) Is complete separation of Br^- and CrO_4^{2-} by fractional precipitation feasible?

Given that K_{sp} of $Ag_2CrO_4 = 1.1 \times 10^{-12}$ and K_{sp} of AgBr = 5.0×10^{-13}.

Suggested Solution:

(a) The required values of $[Ag^+]$ for precipitation to start are

AgBr ppt: $Q_{sp} = [Ag^+][Br^-] = [Ag^+](0.010) = 5.0 \times 10^{-13} = K_{sp}$

$[Ag^+] = 5.0 \times 10^{-11}$ M

Ag_2CrO_4 ppt: $Q_{sp} = [Ag^+]^2[CrO_4^{2-}] = [Ag^+]^2(0.010)$

$= 1.1 \times 10^{-12} = K_{sp}$

$[Ag^+]^2 = 1.1 \times 10^{-10}$ and $[Ag^+] = 1.0 \times 10^{-5}$ M

Because the $[Ag^+]$ required to start the precipitation of AgBr(s) is much less than that for Ag_2CrO_4(s), AgBr(s) precipitates first. As long as AgBr(s) is forming, the silver ion concentration can only slowly approach the value required for the precipitation of Ag_2CrO_4(s).

Example 6 (Continued)

Suggested Solution:

(b) As AgBr(s) precipitates, $[Br^-]$ gradually decreases, and this permits $[Ag^+]$ to increase. When $[Ag^+]$ reaches 1.0×10^{-5} M, precipitation of $Ag_2CrO_4(s)$ begins. To determine $[Br^-]$ at the point at which $[Ag^+] = 1.0 \times 10^{-5}$ M, we use K_{sp} for AgBr and solve for $[Br^-]$.

$$K_{sp} = [Ag^+][Br^-] = (1.0 \times 10^{-5})[Br^-] = 5.0 \times 10^{-13}$$

$$[Br^-] = \frac{5.0 \times 10^{-13}}{1.0 \times 10^{-5}} = 5.0 \times 10^{-8}\ M$$

(c) Before $Ag_2CrO_4(s)$ begins to precipitate, $[Br^-]$ will have been reduced from 1.0×10^{-2} M to 5.0×10^{-8} M. Essentially, all the Br^- will have precipitated from solution as AgBr(s), whereas the CrO_4^{2-} remains in solution. Fractional precipitation is feasible for separating mixtures of Br^- and CrO_4^{2-}.



Class Practice 7

Separating Ions by Fractional Precipitation

$AgNO_3(aq)$ is slowly added to a solution with $[Cl^-]$ = 0.115 M and $[Br^-]$ = 0.264 M . What percent of the Br^- remains unprecipitated at the point at which AgCl(s) begins to precipitate?

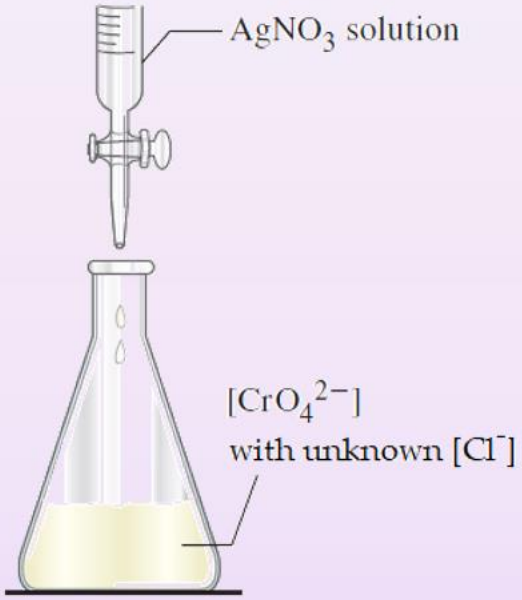
Given that K_{sp} of AgCl = 1.8×10^{-10} and K_{sp} of AgBr = 5.0×10^{-13}.

Precipitation Titration

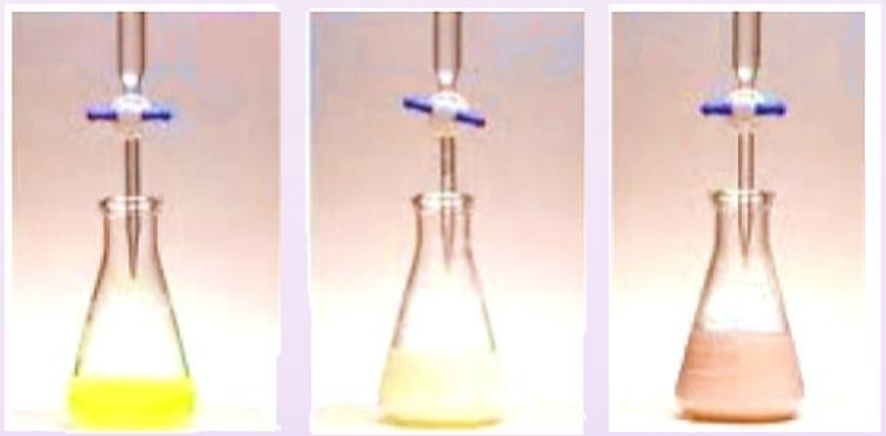
Mohr's Method

- In Mohr's method, the chloride ion concentration of a water sample is determined by ***precipitation titration***. The sample solution is titrated against standard silver nitrate ($AgNO_3$) solution using potassium chromate (K_2CrO_4) as the indicator.

Example: Sample solution with 0.01 M K_2CrO_4 is titrated against 0.01 M $AgNO_3$



	K_{sp}
$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$	1.8×10^{-10}
$Ag_2CrO_4(s) \rightleftharpoons 2\,Ag^+(aq) + CrO_4^{2-}(aq)$	1.1×10^{-12}



initial | before the end point | the end point

Precipitation Titration

Limitation

- Mohr's method should be carried out in a pH range of 6.5–9.0. Otherwise, some ***side reactions*** may affect the titration results.

- If the **pH is too high**, the silver ions may react with the hydroxide ions, forming silver oxide precipitate.

$$Ag^+(aq) + OH^-(aq) \rightarrow AgOH(s)$$
$$2AgOH(s) \rightarrow Ag_2O(s) + H_2O(l)$$

- If the **pH is too low**, the chromate ions (CrO_4^{2-}) will be converted to hydrogen chromate ions ($HCrO_4^-$) or *dichromate ions* ($Cr_2O_7^{2-}$).

$$CrO_4^{2-}(aq) + H^+(aq) \rightleftharpoons HCrO_4^-(aq)$$
$$2\ CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$$

- In each of the above cases, **more Ag^+ are consumed** before the end point is reached. The titration results would give a ***higher Cl^- concentration*** than expected.

Solubility and pH

- The pH of a solution can significantly affect the solubility of a salt. This is especially true when the anion of the salt is the conjugate base of a weak acid or the base OH^- itself.

- Example: Milk of magnesia, an aqueous suspension of $Mg(OH)_2(s)$, is a popular antacid. Hydroxide ions from the dissolved magnesium hydroxide react with hydronium ions (in gastric juice) to form water.

$$Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\,OH^-(aq)$$
$$OH^-(aq) + H_3O^+(aq) \longrightarrow 2\,H_2O(l)$$



Milk of magnesia, an aqueous suspension of $Mg(OH)_2$ (s).

Solubility and pH

$$Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\,OH^-(aq) \quad (1)$$

$$OH^-(aq) + H_3O^+(aq) \longrightarrow 2\,H_2O(l) \quad (2)$$

- According to Le Châtelier's principle, reaction (*1*) would shift to the right because additional $Mg(OH)_2$ would dissolve to replace OH^- ions consumed by the neutralisation (*2*). The overall net ionic equation is:

$$Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + \cancel{2\,OH^-(aq)} \qquad K_{sp} = 1.8 \times 10^{-11}$$

$$\cancel{2\,OH^-(aq)} + 2\,H_3O^+(aq) \rightleftharpoons 4\,H_2O(l) \qquad K' = 1/K_w{}^2 = 1.0 \times 10^{28}$$

$$Mg(OH)_2(s) + 2\,H_3O^+ \rightleftharpoons Mg^{2+}(aq) + 4\,H_2O(l) \quad (3)$$

$$K = K_{sp}/K_w{}^2 = 1.8 \times 10^{-11} \times 1.0 \times 10^{28} = 1.8 \times 10^{17}$$

- The large value of *K* for reaction (*3*) indicates that a ***greater*** amount of $Mg(OH)_2$ will react (dissolve) in acidic solution than in pure water.

- Similarly, other slightly soluble basic solutes (e.g. $CaCO_3$, MgF_2, and CaC_2O_4) also dissolve to a greater extent in acidic solutions.

Solubility and pH

- For slightly soluble basic solutes, we can write overall net ionic equations for solubility equilibria and corresponding K values based on K_{sp} for the solutes and K_a for the conjugate acids of the anions.

Example 1: CaF_2

$$CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2\ F^-(aq) \qquad K_{sp} = 5.3 \times 10^{-9}$$

$$2\ F^-(aq) + 2\ H_3O^+(aq) \rightleftharpoons 2\ HF\ (aq) + 2\ H_2O\ (l) \qquad K' = 1/(K_a)^2 = 2.3 \times 10^5$$

$$CaF_2(s) + 2H_3O^+(aq) \rightleftharpoons Ca^{2+}(aq) + 2\ HF\ (aq) + 2\ H_2O\ (l)$$

$$K = K_{sp}/K_a^{\,2} = 1.22 \times 10^{-3}$$

Example 2: $CaCl_2$

$$CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2\ Cl^-(aq) \qquad K_{sp} = 5.3 \times 10^{-9}$$

$$2\ Cl^-(aq) + 2\ H_3O^+(aq) \rightleftharpoons 2\ HCl\ (aq) + 2\ H_2O\ (l) \qquad K' = 1/(K_a)^2 = 5.9 \times 10^{-13}$$

$$CaCl_2(s) + 2H_3O^+(aq) \rightleftharpoons Ca^{2+}(aq) + 2\ HCl\ (aq) + 2\ H_2O\ (l)$$

$$K = K_{sp}/K_a^{\,2} = 3.14 \times 10^{-21}$$

Solubility and pH

Example 3: CaC_2O_4

$$CaC_2O_4\,(s) \rightleftharpoons Ca^{2+}\,(aq) + C_2O_4^{2-}(aq) \qquad K_{sp} = 2.27 \times 10^{-9}$$

$$C_2O_4^{2-}(aq) + H_3O^+\,(aq) \rightleftharpoons HO_2CCO_2^-\,(aq) + H_2O\,(l) \qquad K' = 1/K_{a1} = 1.70 \times 10^2$$

$$HO_2CCO_2^-(aq) + H_3O^+\,(aq) \rightleftharpoons HO_2CCO_2H\,(aq) + H_2O\,(l) \qquad K'' = 1/K_{a2} = 1.55 \times 10^4$$

$$CaC_2O_4\,(s) + 2\,H_3O^+\,(aq) \rightleftharpoons Ca^{2+}\,(aq) + HO_2CCO_2H\,(aq) + 2\,H_2O\,(l)$$

$$K = K_{sp}/(K_{a1}\,K_{a2}) = 5.97 \times 10^{-4}$$

Example 4: $CaSO_4$

$$CaSO_4\,(s) \rightleftharpoons Ca^{2+}\,(aq) + SO_4^{2-}(aq) \qquad K_{sp} = 9.1 \times 10^{-6}$$

$$SO_4^{2-}(aq) + H_3O^+\,(aq) \rightleftharpoons HSO_4^-\,(aq) + H_2O\,(l) \qquad K' = 1/K_{a1} = 1.58 \times 10^{-3}$$

$$HSO_4^-(aq) + H_3O^+\,(aq) \rightleftharpoons H_2SO_4\,(aq) + H_2O\,(l) \qquad K'' = 1/K_{a2} = 9.77 \times 10^2$$

$$CaSO_4\,(s) + 2\,H_3O^+\,(aq) \rightleftharpoons Ca^{2+}\,(aq) + H_2SO_4\,(aq) + 2\,H_2O\,(l)$$

$$K = K_{sp}/(K_{a1}\,K_{a2}) = 1.41 \times 10^{-6}$$

Example 7

Determining Whether a Precipitate Will Form in a Solution in Which There Is Also an Ionization Equilibrium

Should $Mg(OH)_2$ (s) precipitate from a solution that is 0.010 M $MgCl_2$ and also 0.10 M NH_3 ?
$K_{sp}\,[Mg(OH)_2] = 1.8 \times 10^{-11}$; $K_b\,[NH_3] = 1.8 \times 10^{-5}$

Suggested Solution:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq) \qquad K_b = 1.8 \times 10^{-5}$$

Let the change of $[NH_3]$ be x,
such that at equilibrium, $[NH_4^+] = x$ M, $[OH^-] = x$ M, $[NH_3] = (0.10 - x) \approx 0.10$ M

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{x \cdot x}{0.10} = 1.8 \times 10^{-5}$$

$$x^2 = 1.8 \times 10^{-6} \qquad x = 1.3 \times 10^{-3} \qquad [OH^-] = x\,\text{M} = 1.3 \times 10^{-3}\,\text{M}$$

$$Q_{sp} = [Mg^{2+}][OH^-]^2 = (1.0 \times 10^{-2})(1.3 \times 10^{-3})^2$$
$$= 1.7 \times 10^{-8} > K_{sp} = 1.8 \times 10^{-11}$$

As a result, precipitation occurs.



Class Practice 8

Determining Whether a Precipitate Will Form in a Solution in Which There Is Also an Ionization Equilibrium

Should $Mg(OH)_2(s)$ precipitate from a solution that is 0.010 M $MgCl_2(aq)$ and also 0.10 M $CH_3COO^-Na^+$?

$K_{sp}\,[Mg(OH)_2] = 1.8 \times 10^{-11}$; $K_a\,[CH_3COOH] = 1.8 \times 10^{-5}$

Example 8

Controlling an Ion Concentration, Either to Cause Precipitation or to Prevent It

What $[NH_4^+]$ must be maintained to prevent precipitation of $Mg(OH)_2$ (s) from a solution that is 0.01 M $MgCl_2$ and 0.10 M NH_3 ?

$K_{sp}\,[Mg(OH)_2] = 1.8 \times 10^{-11}$; $K_b\,[NH_3] = 1.8 \times 10^{-5}$

Suggested Solution:

$$[Mg^{2+}][OH^-]^2 = (1.0 \times 10^{-2})[OH^-]^2 = 1.8 \times 10^{-11}$$
$$[OH^-]^2 = 1.8 \times 10^{-9}$$
$$[OH^-] = 4.2 \times 10^{-5}\,M$$

Next let's determine what $[NH_4^+]$ must be present in 0.10 M NH_3 to maintain $[OH^-] = 4.2 \times 10^{-5}$ M.

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq) \qquad K_b = 1.8 \times 10^{-5}$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{[NH_4^+](4.2 \times 10^{-5})}{0.10} = 1.8 \times 10^{-5}$$

$$[NH_4^+] = \frac{0.10 \times 1.8 \times 10^{-5}}{4.2 \times 10^{-5}} = 0.043\,M$$

To keep the $[OH^-]$ at 4.2×10^{-5} M *or less*, and thus prevent the precipitation of $Mg(OH)_2$(s), $[NH_4^+]$ should be maintained at 0.043 M or *greater*.



Class Practice 9

Controlling an Ion Concentration, Either to Cause Precipitation or to Prevent It

What minimum [NH_4^+] must be present to prevent precipitation of $Mn(OH)_2(s)$ from a solution that is 0.0050 M $MnCl_2$ and 0.025 M NH_3?

K_{sp} [$Mn(OH)_2$] = 1.9×10^{-13}; pK_b [NH_3] = 4.74